

# Qihao Shao

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## Research Interests

- LLM agents, Agent Memory, and tool-use agents.
- Uncertainty estimation, model calibration, and reliable decision evaluation.
- Benchmark design, long-horizon task evaluation, and engineering evaluation systems.

## Education

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|----------------------------------------------------------------|-------------------|
| Dalian No. 24 High School                                      | 2021.09 – 2024.06 |
| High School                                                    |                   |
| University of Chinese Academy of Sciences, Cyberspace Security | 2024.09 – Present |
| Undergraduate                                                  |                   |

## Research Experience

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| Agent Memory Benchmark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026.06 – Present |
| Causal Ripple Effect in Agent Memory   Research in Progress                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |
| <ul style="list-style-type: none"><li>• Surveyed HorizonBench, STALE, PersonaMem-v2, and LongMemEval, focusing on task formulation, data construction, and evaluation blind spots.</li><li>• Summarized data construction patterns for long-term memory evaluation, including state-first generation, old/new observation pairs, and pre-evolution hard negatives.</li><li>• Contributed to an Agent Memory evaluation framework covering preference evolution, stale-memory suppression, over-personalization, and premise resistance.</li><li>• Organized evaluation signals such as evolved-static gap, old-preference distractor rate, and scenario applicability judgment.</li></ul> |                   |

## Projects

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| Qwen-vLLM Inference Service Optimization                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026.06 – 2026.07 |
| Pilot Cup, National College Student Computer System Capability Competition   Team Lead                                                                                                                                                                                                                                                                                                                                                                                                               |                   |
| <ul style="list-style-type: none"><li>• Built a Qwen inference optimization branch on top of vLLM under competition constraints, focusing on long-context prefill, decode, KV cache behavior, sampling hot paths, and custom operator overhead.</li><li>• Used Hipprof and runtime artifacts to analyze attention paths, TTFT, TPOT, and throughput bottlenecks.</li><li>• Improved official benchmark throughput by 143% for 16–32K, 75% for 8–16K, and 49% for 4–8K; final rank: 57/346.</li></ul> |                   |

## Skills

|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
| Programming   | Python, C/C++, JavaScript                                                             |
| Deep Learning | PyTorch, Transformers, Hugging Face, vLLM                                             |
| Agent / LLM   | Prompt Engineering, Tool-use Agent, RAG, Memory Mechanism, LangChain                  |
| Evaluation    | Benchmark Design, Metric Design, Ablation Study, Error Analysis, Calibration Analysis |
| Engineering   | Git, Linux, Docker, Conda, LaTeX                                                      |
| Data          | NumPy, Pandas, Matplotlib, JSON, SQLite                                               |

## Honors

- Outstanding Communist Youth League Member, University of Chinese Academy of Sciences, 2025.
- Third-class Undergraduate Scholarship, University of Chinese Academy of Sciences, 2025.